Introduction to Raspberry Pi

1. **General Overview:**
   - What is a Raspberry Pi 4? Describe its primary purpose and typical applications.
2. **Hardware Specifications:**
   - List the key hardware specifications of the Raspberry Pi 4, including the CPU, GPU, RAM options, and available ports.
3. **Operating System:**
   - What operating system(s) can run on the Raspberry Pi 4? Provide a brief description of at least two different OS options.
4. **Networking:**
   - Describe the networking capabilities of the Raspberry Pi 4. Include details on both wired and wireless options.
5. **Power Supply:**
   - What are the power requirements for the Raspberry Pi 4? Mention the recommended power supply specifications.
6. **GPIO Pins:**
   - What are GPIO pins, and how are they used in the Raspberry Pi 4? Provide an example of a simple project that utilizes GPIO pins.
7. **Storage:**
   - What storage options are available for the Raspberry Pi 4? Explain how you can expand the storage capacity.
8. **Software Development:**
   - List some programming languages supported by the Raspberry Pi 4. Provide an example of a software development tool or environment commonly used.
9. **Peripherals and Accessories:**
   - Name three peripherals or accessories that can enhance the functionality of the Raspberry Pi 4. Briefly explain the use of each.
10. **Community and Support:**
    - How can students find help and resources for Raspberry Pi 4 projects? Mention at least two community forums or official resources.

Reflection:

- Based on your gathered technical details, how do you envision using the Raspberry Pi 4 in your future projects or coursework? Provide a brief reflection.